

Ayon Kumar Bhowmick Ovi

B.Sc in Computer Science and Engineering, AIUB

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CAREER GOAL

To begin my career in a challenging role where I can apply my knowledge of software development, machine learning, and data analysis to solve real-world problems. I want to work with experienced professionals, learn modern tools and industry practices, and take on responsibilities that strengthen my technical and problem-solving skills. My long-term goal is to grow into a skilled and dependable engineer who contributes to useful, high-quality products and to the success of the organization.

EDUCATION

American International University Bangladesh (AIUB) — **Dhaka, Bangladesh**
B.Sc in Computer Science and Engineering | CGPA: 3.71/4.00 (Current) — *Jan 2023 - Present*

Dhaka Imperial College — **Dhaka, Bangladesh**
Higher Secondary Certificate — *2019-2021, GPA: 5.00*

National Ideal School — **Dhaka, Bangladesh**
Secondary School Certificate — *2017-2019, GPA: 4.78*

TECHNICAL SKILLS

Programming: Python, C++, C#, Java, JavaScript, PHP
Web & Software: HTML, CSS, .NET, Windows Forms, MySQL,SQL, Microsoft SQL Server
AI/ML: PyTorch, Pandas, NumPy, Scikit-learn
Testing & QA: Manual & API Testing (Postman), Test Case Design, Selenium, SQA
Tools & OS: Git, GitHub, VS Code, Jupyter, Google Colab, Windows, Linux

PROJECTS

Café Shop Management System *[C#, SQL Server]*
github.com/AyonBhowmick/Cafe_Shop_Management_System

Built a café management system to handle daily operations and customer orders, using C# and Microsoft SQL Server with role-based access for Admin and Cashier users, enabling centralized management of products, staff accounts, payments, receipts, and sales records.

Fund Raiser Management System *[Oracle SQL]*
github.com/AyonBhowmick/Fund_Raiser_Management_System

Designed a centralized relational database to replace scattered donor spreadsheets that caused duplicate records and slow reporting. Linked events, donors, donations, organizations, and teams in a schema normalized to 3NF, and wrote joins, subqueries, and views to report total donations and top donors.

University Management System *[Java, Swing, File I/O]*
github.com/AyonBhowmick/University_Management_System

Built a desktop system to manage student, faculty, and administrative records, using Java Swing with role-based login and File I/O for persistent storage, supporting tuition and scholarship fee management.

SkillSwap Connect *[PHP, MySQL, HTML, CSS, JavaScript]*
github.com/AyonBhowmick/SkillSwap-Connect

Built a web platform to connect learners and mentors for skill exchange and paid courses, using PHP, MySQL, and JavaScript with role-based features for Learners, Mentors, and Admins, delivering course requests, messaging, progress tracking, and certificate generation.

Virus Invaders *[C++, OpenGL, GLUT]*
github.com/AyonBhowmick/VIRUS_INVADERS

Built a 2D game in C++ and OpenGL/GLUT with collision detection, health and score tracking, and five progressive levels with dynamic weather effects.

RESEARCH INTERESTS

Data Analytics, Machine Learning, Business Intelligence, Deep Learning, Natural Language Processing (NLP), Data Mining, Computer Vision, Data Science.

PUBLICATIONS

[1] D. P. Karmakar, **A. K. B. Ovi**, T. B. Jenny, and M. M. Hasan, "Daily Peak Demand Forecasting in Bangladesh: A Leakage-Aware Comparison of Machine Learning, Deep Learning, and a Hybrid RF-Naive Model," *2026 International Conference on Power, Electronics, Communications, Computing, and Intelligent Infrastructure (PECCII)*, Pabna, Bangladesh, 2026, pp. 1-6, doi: 10.1109/PECCII70991.2026.11661966.

[2] **A. K. B. Ovi**, S. K. B. Shanto, and B. K. Bhawmick, "An Explainable Stacking Ensemble Framework with SMOTE for Loan Approval Prediction: A CIBIL-Centric Interpretability Analysis," *2026 IEEE 8th International Conference on Sustainable Technologies for Industry 5.0 (STI)*, Dhaka, Bangladesh, 2026, accepted for oral presentation (to appear).

RESEARCH IN PROGRESS

Papers Under Peer Review

- **Leakage-Aware Diabetes Prediction Using Mutual Information and Explainable Machine Learning**
  A. K. B. Ovi and B. K. Bhawmick, submitted to the *29th International Conference on Computer and Information Technology (ICCIT)*, 2026.
- **Chilli Leaf Disease Classification Using a ConvNeXt-Mamba Hybrid with Explainable AI**
  A. K. B. Ovi, T. B. Jenny, T. Ahmed, R. Mahmud, and S. A. Tasnim, submitted to the *29th International Conference on Computer and Information Technology (ICCIT)*, 2026.
- **Explainable Machine Learning for Early Dengue Detection Using Routine Blood Parameters: A CTGAN-Balanced Stacking Framework**
  A. K. B. Ovi, S. K. B. Shanto, B. K. Bhawmick, and D. J. Gomes, submitted to the *5th International Conference on Robotics, Electrical and Signal Processing Techniques (ICREST)*, 2027.

Manuscripts in Preparation

- **Software Defect Prediction Using Static Code Metrics: A Pair-Aware and Explainable Machine Learning Approach**
  A. K. B. Ovi, H. Saha, T. Rahaman, M. S. Jahan, S. J. Liza, and M. A. A. Mahmud.

ACHIEVEMENTS & CERTIFICATIONS

- **Cisco Networking Academy** - Completed networking-related training and certification.
- **IEEE SPAVe 7.0** - Participated in the IEEE Student Professional Awareness Venture (SPAVe) 7.0 organized by the IEEE AIUB Student Branch.

PROFESSIONAL MEMBERSHIP

- **Student Member**, IEEE AIUB Student Branch 2026 - Present

REFERENCES

Dipta Justin Gomes
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American International University-Bangladesh (AIUB)
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Mukta Das
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